

Thursday	5	12	19	26	
Friday	6	13	20	27	2
Saturday	7	14	21	28	0
Sunday	1	8	15	22	2
Week	18/23	19	20	21	22

APR 2022

thursday
(111-254) Week 17

21

9.00 Java ⇒ Java is high level
9.30 language
10.00 Microsystem, developed by Sun
10.30
11.00

11.30 ► Main strength of Java's that
12.00 it follows write once
12.30 run anywhere? principle
1.00

1.30 that means compiled Java code
2.00 can be run on any device
2.30 that has JVM installed.
3.00

3.30 ► Java allow to build secure
4.00 , scalable , maintainable applications.
4.30

5.00 moreover Java's platform independent
5.30 language • its OO nature allow
6.00 to write clean and
reusable code.

APR 2022

22

friday

(112-253) Week 17

A	Monday	4	11	18	25
P	Tuesday	5	12	19	26
R	Wednesday	6	13	20	27
	Thursday	7	14	21	28
2	Friday	1	8	15	22
0	Saturday	2	9	16	23
2	Sunday	3	10	17	24
Week		14	15	16	17

JDK $\Rightarrow$ is software development environment that offers and library needed for developing Java based application.

it include * Javac
* JRE

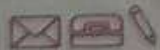
JRE $\Rightarrow$ is software

package that provide environment required to run Java application

JVM $\Rightarrow$ JVM is part of

JRE that's responsible for executing Java Byte code.

It allow Java programs



9.00 to be
9.30 running
10.00 code
10.30 extension
11.00
11.30
12.00

platform-independent by
the same compiled
on any OS have
.class

12.00 Byte Code ⇒ Byte code's
1.00
1.30 Compiled version of your
2.00 code that's Not readable
2.30 By human but can be
3.00 interpreted on executed by
virtual machine like JVM!

★ STEPS ★

- Write the Code in .Java files
- Compile code with javac which's part of JDK
- Run the Java code ⇒
 - * Java command invoke JVM
 - * JVM reads Bytecode in .class file and then convert into Machine Code

APR 2022

27

wednesday

(17-248) Week 18

8	Wednesday	6	13	20	27
7	Thursday	7	14	21	28
6	Friday	8	15	22	29
5	Saturday	9	16	23	30
4	Sunday	10	17	24	
Week		14	15	16	17

9:00 << Left Shift $\Rightarrow 9 \ll 1$

9:30

10:00

10:30

11:00

11:30

12:00

12:30

1:00

1:30

★ Think we have ask $9 \ll 2$
 it mean at left side add
 that zero's

3:30

4:00

4:30

5:00

5:30

6:00

9 binary $\Rightarrow 1001$

and $9 \ll 2 \Rightarrow 100100 \Rightarrow 36$

16 8 4 2 1

32 16 8 4 2 1

Monday	31	3	10	17	24	31
Tuesday		4	11	18	25	
Wednesday		5	12	19	26	
Thursday		6	13	20	27	
Friday		7	14	21	28	
Saturday	1	8	15	22	29	
Sunday						
Week	18/3	19	20	21	22	

APR 2022
thursday 28
(118-247) Week 18

RIGHT SHIFT $\Rightarrow$ shift the
binary value to right side

Think $9 \gg 2$

9 binary $\Rightarrow$ 1001

$9 \gg 2 \Rightarrow$ 10 Binary

2 decimal value

01 will think wrap
after shift

Monday	31	3	10	24	A
Tuesday		4	11	18	Y
Wednesday		5	12	19	—
Thursday		6	13	20	2
Friday		7	14	21	0
Saturday	1	8	15	22	2
Sunday					
Week	18/01	19	20	21	22

APR 2022

tuesday

(116-249) Week 18

26

9 1 0 0 1
 10 1 0 1 0
 AND
 (8) 1 0 0 0

Bitwise XOR $\Rightarrow$ Both Bit same $\Rightarrow 0$
 otherwise $\Rightarrow 1$

9 1 0 0 1
 10 1 0 1 0 XOR
 (3) 0 0 1 1

Bitwise complement $\sim \Rightarrow$

: inverts all the bits

9 $\Rightarrow$ 1001 $\sim \Rightarrow$ 0110

APR 2022

25

monday

[115-250] Week 18

A	Monday	4	11	18	25
P	Tuesday	5	12	19	26
E	Wednesday	6	13	20	27
T	Thursday	7	14	21	28
F	Friday	1	8	15	22
S	Saturday	2	9	16	23
S	Sunday	3	10	17	24
Week		14	15	16	17

Binary operators

are those operators that work on bit level.

Bitwise OR $\Rightarrow$ it will go on each corresponding bit if any 1 bit found then 1.

$$\begin{array}{r}
 9 \Rightarrow 1001 \\
 10 \Rightarrow 1010 \\
 \hline
 11
 \end{array}$$

Bitwise AND $\Rightarrow \&$

return 1 only if both corresponding bit are 1